
Conformal Prediction with Conditional Coverage and Non-Exchangeability Guarantees

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1 Problem Statement

Traditional methods for constructing prediction intervals often rely on specific assumptions about the distribution of forecasting errors, such as normality or homoskedasticity. While these methods, yield precise intervals when their assumptions hold true, their validity breaks down when these conditions are violated. Conformal prediction has emerged as an appealing framework in modern predictive inference. Given i.i.d. data (X_i, Y_i) , conformal methods construct prediction sets $C(X_{n+1})$ satisfying finite sample **marginal** coverage,

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha,$$

without making any distributional assumptions. This means that on average over all data points, the prediction intervals contain the outcome at the desired coverage rate. This distribution-free property has led to the understanding that conformal prediction is 'distribution-free'.

However, for many applications we are more interested in conditional coverage. For example, in personalised medicine, a clinician might seek a prediction range for the effectiveness of a drug based on the *specific* characteristics of the patient (e.g. age, symptoms, family history), instead of relying on a generic range printed the back of the packaging that is valid only for the "average" patient. The objective of interest is therefore **conditional coverage**

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha \quad \text{for all } x.$$

Unfortunately, a negative result by Vovk (2012); Lei and Wasserman (2014) shows that distribution-free conditional coverage is impossible. This has been interpreted as a fundamental limitation of conformal prediction

The central idea of this project is that this conclusion is misinterpreted. In this work we argue that conformal prediction should not be understood as a distribution-free method, but as a model calibrator. It takes whatever distributional model we choose to encode, and if its assumptions are (approximately) correct, we achieve exact conditional coverage guarantees. but at the benefit that marginal guarantees are always maintained, no matter how poor our model.

This project aims to develop a rigorous theory of conformal prediction that:

1. Characterises what is required to achieve conditional coverage.
2. Shows how distributional assumptions can be made explicit and calibrated.
3. Quantifies the exact conditional coverage gap with inaccurate model specifications.
4. Extends to many new areas for conformal prediction.

2 The pivotal role of the non-conformity score

The starting point is the idea of a **conditional pivot**.

A real-valued score $R = r(X, Y)$ is a conditional pivot for a distribution $\mathbb{P}$ on (X, Y) if its conditional distribution does not depend on the covariates:

$$R \mid X = x \sim G \quad \text{for all } x,$$

for some CDF G (not necessarily uniform). If we choose such a score in split conformal prediction, then the calibration scores and the test scores are i.i.d. from G , independent of X . The standard order statistic argument then provides exact finite-sample conditional coverage:

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) = \mathbb{P}(Y_{n+1} \in C(X_{n+1}) \geq 1 - \alpha).$$

This then reframes the classical impossibility result: distribution-free conditional coverage is impossible because there is no universal conditional pivot valid for all distributions. But it motivates, us to choose a non-conformity score that to the best of our ability behaves like a conditional pivot.

The most direct way to construct a pivot is through the conditional CDF of Y given X . Suppose that, for each x , we know the true conditional distribution

$$F_{Y|X}(y \mid x) = \mathbb{P}(Y \leq y \mid X = x)$$

and we define the score

$$R = F_{Y|X}(Y \mid X).$$

By the probability-integral transform, for each x , the conditional distribution of $R \mid X = x$ is uniform(0, 1). Hence R is a conditional pivot, and its distribution is the same for every x , and therefore enjoys finite sample conditional coverage.

Of course in practise we do not have perfect knowledge of $F_{Y|X}$. Instead, we often have an approximate model $\tilde{F}$, estimated on the training sample. The question is then if we use this as our score and apply conformal prediction, how far is the resulting conditional coverage from the target $1 - \alpha$?

Assume that the approximation is uniformly accurate:

$$\sup_{x,y} \left| \tilde{F}(y \mid x) - F_{Y|X}(y \mid x) \right| = \epsilon$$

then we can show that $\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X = x) \geq 1 - \alpha - 2\epsilon$. Proof in the appendix.

This idea has been explored in (Chernozhukov et al., 2021), which argues that if the conditional cdf could be consistently estimated, this provides *asymptotic* conditional coverage guarantees. This analysis however does not include any explicit loss bounds.

One can sharpen this analysis by focusing not on the uniformity, but on the heterogeneity of its conditional distribution across x . This leads to the precise quantification of loss in conditional coverage, which we will cover in the next section.

2.1 Extending the framework to standard non-conformity scores

In many applications, practitioners work not with the full conditional CDF of Y , but with a non-conformity score based on the point forecast. This score, as described in the literature, measures how 'strange' or 'non-conforming' an outcome is relative to the point prediction, hence the name. But its definition remains rather vague. A canonical example is the absolute standardised residual

$$R = r(X, Y) = \frac{|Y - \hat{f}(X)|}{\hat{\sigma}(X)}$$

where $\hat{f}$ is a prediction model of the mean or median, and $\hat{\sigma}(X)$ a scale estimate. A practitioner might choose such a score to adapt to heteroskedasticity and scale large residuals in high variance regions accordingly.

From our distributional perspective, the practitioner may believe that, under a correct model specification, the standardised residual $R \mid X = x$, comes from a distribution that is approximately normal, or at least symmetric and unimodal, with a distribution that does not.

- Mention that the correct interpretation is that it should be a distance measure for the distribution.

3 Monotone invariance

A particularly important and interesting property is its invariance to monotone transformations of the score. Let $R' = \phi(R)$, where $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Denote by

An example illustrates the point. Suppose the true dat-generating mechanism is

$$Y = f(X) + \sigma(X)Z^3$$

where $Z \sim \mathcal{N}(0, 1)$ independent of X . The natural "z-score" in this situation is

$$R = \frac{Y - f(X)}{\sigma(X)} = Z^3$$

The practitioner might nevertheless incorrectly believe that $S \mid X = x$ is standard normal for all x , and the scores conditional CDF is

$$G_x(t) = \mathbb{P}(\Phi(Z^3) \leq t \mid X = x) = \Phi\left((\Phi^{-1}(t))^{1/3}\right).$$

This CDF is not uniform, and does not equal the normal CDF $\Phi(t)$. The practitioner's belief about the distributional shape is wrong. Yet $G_x = G$ for all x as the distribution does not depend on the covariates. Thus it is still a pivot,

This example illustrates the message that what matters for conditional validity is not whether its PIT variable is uniform, or whether we got the exact parametric family right but whether our score is a conditional pivot. It furthermore highlights the attractive modelling property that our model only needs to be approximately correct, and can be incorrect for some global misspecification that equally affects all x .

4 Quantifying the conditional coverage gap

The previous discussion suggests that the right way to measure how far we are from conditional coverage is not by asking how close a PIT variable is to uniform, but by asking how much its conditional distribution varies with X . For this we first need to define what variation means in this context, as variation is commonly measured against a reference point. To do this we measure the distance between the conditional CDF of the non-conformity score against its marginal CDF.

Let $R = R(X, Y)$ be any non-conformity score. For each x , define its conditional CDF

$$G_x(t) = \mathbb{P}(R \leq t \mid X = x)$$

and denote by

$$G(t) = \mathbb{P}(R \leq t) = \mathbb{E}_X[G_X(t)]$$

the marginal CDF of R . If R is a conditional pivot, then $G_x = G$ for all x . Deviations from this quantify the failure of the pivot property. A natural measure of heterogeneity in the conditional PIT distribution is

$$\Delta = \sup_{x,t} |G_x(t) - G(t)|$$

This KS-type distance between the conditional distribution G_x and the mixture G is zero if and only if R has the same distribution for every covariate value.

With these tools we can show that Δ controls the conditional coverage loss in a particularly neat way. Consider again the split conformal prediction procedure applies to the scores $(R_1, \dots, R_n)$. Let $\hat{Q}$ be the empirical $(1 - \alpha)$ -quantile of the calibration scores $R_1, \dots, R_n$. Note that marginal coverage is

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) = \mathbb{P}(R_{n+1} \leq \hat{Q}) = \mathbb{E}[G(\hat{Q})]$$

And that conditional coverage is

$$\mathbb{P}(R_{n+1} \leq \hat{Q} \mid X_{n+1} = x) = \mathbb{E}[G_x(\hat{Q})]$$

where in both the expectation is taken over the calibration data, and hence the randomness of $\hat{Q}$. By definition of Δ we have $G_x(\hat{Q}) \geq G(\hat{Q}) - \Delta$. Hence

$$\mathbb{P}(R_{n+1} \leq \hat{Q} \mid X_{n+1} = x) \geq \mathbb{E}[G(\hat{Q})] - \Delta = \mathbb{P}(R_{n+1} \leq \hat{Q}) - \Delta.$$

Therefore,

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq \mathbb{P}(Y_{n+1} \in C(X_{n+1})) - \Delta \geq 1 - \alpha - \Delta.$$

These tools allow us to reason between marginal and conditional coverage guarantees, and thus shows that the conditional coverage loss is controlled entirely by the variation of the conditional CDF across covariates.

This observation allows us to clarify and refine several points:

First, it justifies the statement that only the variation of the conditional CDFs matter for conditional coverage, not their shape. As long as $G_x = G$ for all x , which makes $\Delta = 0$. In the Z^3 example, $R = \Phi(Z^3)$ has a non-uniform CDF G , but $\Delta = 0$, and conditional coverage remains intact.

Second, it gives a clean way to decompose the effect of misspecification. In the CDF estimation error assumption

$$\sup_{x,y} \left| \tilde{F}(y \mid x) - F_{Y \mid X}(y \mid x) \right| = \varepsilon,$$

and define the PIT-like score $\tilde{R} = \tilde{F}(Y \mid X)$. One can show that this implies

$$\sup_{x,t} |G_x(t) - t| = \varepsilon$$

we can decompose this error into

$$|G_x(t) - t| = |G_x(t) - G(t) + G(t) - t| \leq |G_x(t) - G(t)| + |G(t) - t|.$$

The first term $|G_x(t) - G(t)|$ is exactly the heterogeneity component, while the second term $|G(t) - t|$ is a global miscalibration component that affects all covariates equally. Using the bound $|G_x(t) - t| \leq \varepsilon$ it is easy to prove that $|G(t) - t| \leq \varepsilon$ for all t . Then the decomposition above also equally gives

$$|G_x(t) - G(t)| \leq |G_x(t) - t| + |G(t) - t| \leq \varepsilon + \varepsilon = 2\varepsilon$$

for every x, t . Taking the supremum over x, t shows that

$$\Delta \leq 2\varepsilon.$$

Third, the bound based on Δ aligns with the intuition that conditional coverage is as good as the worst-case discrepancy between the conditional and marginal distributions of the score. In practice, one may estimate Δ empirically, through sliced Kolmogorov-Smirnov tests on R across covariate groups, and adjust the normal level to hedge against the estimated variation $\alpha^* = \alpha + \hat{\Delta}$

5 Multivariate extension and oracle efficient prediction sets

The model based conformal framework extends naturally to multivariate outcomes. This is particularly important because classical conformal methods in absence of distributional assumptions, also places no structure on the dependence relation of the outcomes. The result is that conformal methods produce extremely large prediction sets in high dimensions and would have to resort to bonferroni type corrections, which can be overly conservative.

However by imposing additional structure to the problem and incorporating information about the dependence structure we can significantly decrease the set size, making predictions more informative even when our model is misspecified. For example, if we assume that the residuals come from an elliptical distribution, we might choose to incorporate this into our score function using the Mahalanobis distance as done in Xu et al. (2024).

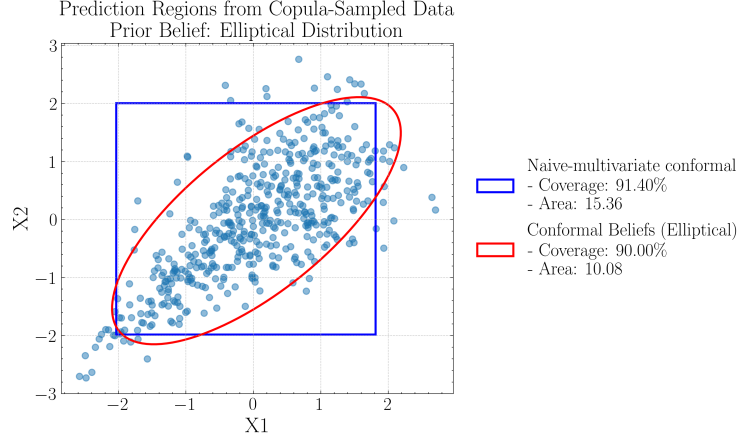


Figure 1: Enter Caption

Furthermore, being explicit about the distributional assumptions we choose to conformally calibrate, if we pin down a set of distributions from the same family. Can choose to calibrate the specific quantile that yield the smallest prediction sets, by selecting the highest density areas. The result is that we maximise efficiency and provide the most informative uncertainty sets.

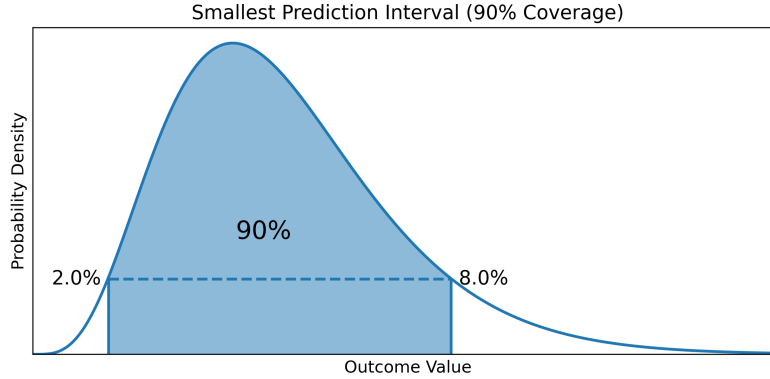


Figure 2: Enter Caption

Aside from achieving conditional coverage another strand of the literature in conformal prediction focuses on achieving the smallest prediction sets, which hence are the most informative, but in a distribution-free sense it is rather unclear how one, but the oracle using the t. in the discussion of achieving the smallest prediction sets does this work in no way claim to provide an original contribution, but attempts to show again that the distributional/model-based perspective is the right way to look at conformal prediction.

Mathematically this section is still severely under-developed

6 Extensions

6.1 Ensemble conformal predictors

Another area of interest is the forecast combination of conformal predictors.

Suppose that instead of a single model we have K different conditional CDF estimators $\tilde{F}^{(k)}(y | x)$, for $k = 1, \dots, K$, each fitted, for example, by a different algorithm or with different hyper-parameters.

For each model we define its uniform CDF error

$$\varepsilon_k = \sup_{x,y} \left| \tilde{F}^{(k)}(y | x) - F_{Y|X}(y | x) \right|$$

Then we combine these models, to form an ensemble conditional CDF by taking a convex combination

$$\tilde{F}^{\text{ens}}(y | x) = \sum_{k=1}^K w_k \tilde{F}^{(k)}(y | x), \quad w_k \geq 0, \quad \sum_{k=1}^K w_k = 1$$

Because each $\tilde{F}^{(k)}(\cdot | x)$ is non-decreasing in y and takes values in $[0, 1]$, their weighted average is again a valid CDF for each x . It can then be shown that the uniform CDF error of the ensemble is bounded by the convex combination of the indivial errors:

$$\begin{aligned} \left| \tilde{F}^{\text{ens}}(y | x) - F_{Y|X}(y | x) \right| &= \left| \sum_{k=1}^K w_k \left(\tilde{F}^{(k)}(y | x) - F_{Y|X}(y | x) \right) \right| \\ &\leq \sum_{k=1}^K w_k \left| \tilde{F}^{(k)}(y | x) - F_{Y|X}(y | x) \right| \leq \sum_{k=1}^K w_k \varepsilon_k \end{aligned}$$

Then taking the supremum over (x, y) shows that the ensemble error satisfies

$$\varepsilon_{\text{ens}} \leq \sum_{k=1}^K w_k \varepsilon_k$$

This demonstrates that the weighted average ensemble performs never worse than the weighted average of its components. In practise, if the individual errors have different signs then averaging can lead to a substantial cancellation and hence ε_{ens} will be strictly smaller than $\sum_k w_k \varepsilon_k$, and possibly even smaller than $\min_k \varepsilon_k$. From the conditional coverage perspective this is precisely what we desire as the ensemble provides a closer approximation to the true pivot. An idea for research purposes is to optimise the weights w_k in order to minimise an empirical proxy for ε_{ens} , and more discussion on this approximation follows in the last section.

The analysis above exploits the linearity of the CDF and that we explicitly model $F_{Y|X}$ (or potentially the CDF of the non-conformity scores). However in the more general framework for generalised non-conformity scores, where we start from arbitrary scores $R = r(X, Y)$ and quantify conditonal coverage loss via

$$\Delta = \sup_{x,t} |G_x(t) - G(t)|$$

there is no generalisation for simply averaging the scores $R^{(k)}$. However, if we are willing to randomise, then an ensemble can offer guarantees similar to the proof above. The idea is to randomise which score function we use for the sample or observation.

Suppose you have K different nonconformity scores:

$$R^{(k)} = r^{(k)}(X, Y) \quad \text{for } k = 1, \dots, K.$$

Each of them, if used alone has its own heterogeneity measure

$$\Delta_k = \sup_t \left| G_x^{(k)}(t) - G^{(k)}(t) \right|$$

An ensemble score is just another score

$$R^{\text{ens}} = r^{\text{ens}}(X, Y)$$

built out of the $R^{(1)}, \dots, R^{(K)}$ (and possibly X , to make the aggregation covariate dependent).

Consider fixed weights $w_1, \dots, w_K \geq 0$ with $\sum_k w_k = 1$. Now draw an index $J \in \{1, \dots, K\}$ with $\mathbb{P}(J = k) = w_k$, independently of all data, and define the ensemble score $R^{\text{ens}} = R^{(J)}$. Then conditioned on $X = x$,

$$G_x^{\text{ens}}(t) = \mathbb{P}(R^{\text{ens}} \leq t | X = x) = \sum_{k=1}^K w_k \mathbb{P}(R^{(k)} \leq t | X = x) = \sum_{k=1}^K w_k G_x^{(k)}(t)$$

and marginally

$$G^{\text{ens}}(t) = \sum_{k=1}^K w_k G^{(k)}(t).$$

Therefore

$$\Delta_{\text{ens}} = \sup_{x,t} |G_x^{\text{ens}}(t) - G^{\text{ens}}(t)| \leq \sum_{k=1}^K w_k \sup_{x,t} |G_x^{(k)}(t) - G^{(k)}(t)| \leq \sum_{k=1}^K w_k \Delta_k$$

As a result we find that the randomisation procedure performs never worse than the worst base score, in terms of conditional coverage loss. Furthermore if the deviations $G_x^{(k)}(t) - G^{(k)}(t)$ have different signs, then the weighted combination can strictly decrease Δ .

In the discussion if this also translates to smaller prediction sets, we have to refine what this means. Generally, speaking it is not meaningful to measure the average prediction set size, as we could always choose to undercover hard predictions with great uncertainty and compensate by over-covering easy predictions. The result would be a small average set size but this would not yield a satisfactory result. Instead, we focus ourselves on the prediction set size when we require conditional coverage. Here we are able to make such claims, as the idea that $\Delta_{\text{ens}} \leq \sum_k w_k \Delta_k$ then we know that if we require the same α for each x , then conditional coverage is closer to nominal, meaning that if we require the same conditional level $1 - \alpha_0$ everywhere, then we can use for the baseline $\alpha_{\text{base}} \dots \alpha_{\text{ens}} > \alpha_{\text{base}}$. Then if we assume that the prediction sets are nested, a larger alpha results in a smaller set.

6.2 Non-exchangeable guarantees

So far, our analysis has been in the i.i.d regression setting, where (X_i, Y_i) are exchangeable and split conformal delivers the marginal guarantee that we extend to conditional coverage loss via the heterogeneity measure $\Delta = \sup_{x,t} |G_x(t) - G(t)|$. However recent literature has attempted to extend conformal prediction to the time-series setting where these assumptions do not hold, see for example (Gibbs and Candes, 2021).

Interestingly with the tools we have developed, while our Δ , definition of heterogeneity was the main work horse for conditional coverage guarantees in the i.i.d. setting. We benefit from our alternative ε approach, that measured the distance in conditional CDFs. as fascinatingly this approach does not require to assume that the data is i.i.d. and it extends to non-exchangeable settings. The key idea is that, in a sequential environment, rather than conditioning on X , we condition on the past filtration. Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $F_t = \sigma(Y_1, \dots, Y_{t-1})$. At each time t , we consider the conditional CDF

$$F_t(y) = \mathbb{P}(Y_t \leq y \mid F_{t-1}),$$

Proposition 1 (Time-series conditional coverage under correct prior) *Let $\{Y_t\}_{t=1}^{T+1}$ be a time-series adapted to F_{t-1} , and suppose we have the true conditional CDFs $F_t(y) = \mathbb{P}(Y_t \leq y \mid F_{t-1})$. Define $R_t = F_t(Y_t)$ and let $\hat{Q}_T$ be the $\lceil (1 - \alpha)(T + 1) \rceil$ -order statistic of $R_1, \dots, R_T$. Then the prediction set*

$$C(F_T) = \{y : F_{T+1} \leq \hat{Q}_T\}$$

satisfies

$$\mathbb{P}(Y_{T+1} \in C(F_T) \mid F_T) \geq 1 - \alpha.$$

As before, we can relax the assumption of a perfectly specified prior and obtain a sharp finite-sample lower bound. Suppose that instead of the true F_t we have approximate conditional CDFs $\tilde{F}_t$, adapted to F_{t-1} and satisfying a uniform sup-norm error bound

$$\sup_{1 \leq t \leq T+1} \sup_{y \in \mathbb{R}} |\tilde{F}_t(y) - F_t(y)| = \varepsilon$$

Then following the same procedure, on the approximate scores $\tilde{R}_t$ yields

Proposition 2 (Approximate prior in time series) *Under the assumptions above,*

$$\mathbb{P}(Y_{T+1} \in \tilde{C}(F_T) \mid F_T) \geq 1 - \alpha - 2\varepsilon.$$

This is a genuine conditional guarantee, and suggests that we are doing the right thing when we apply conformal procedures in the time-series setting.

This result combined with the monotonic invariance, is particularly interesting for example for GARCH models, which place strong distributional assumptions and dynamics on its process. Furthermore the monotone variance property is attractive as there is reasonable debate about the shape of the residual distribution, but any global misspecification such as a systematic under estimation of variance or failure to capture fat tails, can be absorbed into the framework.

6.3 Group conformal+

- Bubble up the idea to choose alternative alpha level for conditional coverage by estimating delta, but allows calibration over all samples. not just in group

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Theorem .1 (Finite-sample conditional coverage under correct full-distributional prior)

Given i.i.d. observations (X_i, Y_i) , $i = 1, \dots, n + 1$, and a correct prior for the conditional distribution of $Y \mid X$, through $F_{Y|X}(y \mid x) = \mathbb{P}(Y \leq y \mid X = x)$, define

$$R_i = F_{Y|X}(Y_i \mid X_i), \quad i = 1, \dots, n + 1.$$

Then, if split-conformal calibration is applied to the transformed values R_i by taking the order-statistic $\hat{Q}_R = R_{(\lceil (1-\alpha)(|\mathcal{D}_{cal}|+1) \rceil)}$, the resulting upper prediction band

$$C(x) = \{y : F_{Y|X}(y \mid x) \leq \hat{Q}_R\}$$

satisfies, for every $x \in \mathcal{X}$,

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha.$$

Proof: By the probability-integral transform and correctness of the prior, each $R_i = F_{Y|X}(Y_i \mid X_i) \sim \text{Unif}(0, 1)$. Hence, by the deterministic transformation of independent pairs (X_i, Y_i) , the $\{R_1, \dots, R_{n+1}\}$ are i.i.d. Uniform(0, 1), and exchangeability implies the standard split conformal result

$$\mathbb{P}(R_{n+1} \leq \hat{Q}_R) \geq 1 - \alpha.$$

But since the $\{R_i\}$ are identically distributed and independent of the covariates, for any fixed x , $\mathbb{P}(R_{n+1} \leq \hat{Q}_R \mid X_{n+1} = x) = \mathbb{P}(R_{n+1} \leq \hat{Q}_R)$, so in fact $\mathbb{P}(R_{n+1} \leq \hat{Q}_R \mid X_{n+1} = x) \geq 1 - \alpha$ for all x . Finally, setting $\hat{Q}_Y(x) = \inf\{y : F_{Y|X}(y \mid x) \geq \hat{Q}_R\}$, so that that $\{Y_{n+1} \leq \hat{Q}_Y(X_{n+1})\} = \{F_{Y|X}(Y_{n+1} \mid X_{n+1}) \leq \hat{Q}_R\}$, maps the coverage bound back to the Y -space. Concluding that for every covariate $x \in \mathcal{X}$,

$$\mathbb{P}(Y_{n+1} \leq \hat{Q}_Y(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha.$$

□

Theorem .2 (Lower-bound Approximate Distributional Beliefs) Let $\{(X_i, Y_i)\}_{i=1}^n$ and (X_{n+1}, Y_{n+1}) be i.i.d. from some distribution P on $\mathcal{X} \times \mathbb{R}$. Fix a miscoverage level $\alpha \in (0, 1)$ and let

$$F_{Y|X}(y | x) = \mathbb{P}\{Y \leq y | X = x\}$$

be the true conditional CDF. Suppose we have an approximate CDF $\tilde{F}(y | x)$ satisfying

$$\sup_{x \in \mathcal{X}} \sup_{y \in \mathbb{R}} |\tilde{F}(y | x) - F_{Y|X}(y | x)| \leq \varepsilon.$$

Define the “conformal scores”

$$\tilde{R}_i = \tilde{F}(Y_i | X_i), \quad i = 1, \dots, n+1,$$

and let

$$m = \lceil (1 - \alpha)(n+1) \rceil, \quad \hat{Q} = \tilde{R}_{(m)},$$

the m -th order statistic of the $\tilde{R}_i$ over the calibration sample $i = 1, \dots, n$. Then the one-sided conformal prediction set

$$C_n(x) = \{y \in \mathbb{R} : \tilde{F}(y | x) \leq \hat{Q}\}$$

satisfies, for **every** covariate x , the finite-sample **conditional** coverage bound

$$\mathbb{P}\{Y_{n+1} \in C_n(x) \mid X_{n+1} = x\} \geq 1 - \alpha - 2\varepsilon.$$

Proof:

(1) By assumption, for every i and every realization (X_i, Y_i) , the true scores $R_i = F_{Y|X}(Y_i | X_i)$ and the approximate scores $\tilde{R}_i = \tilde{F}(Y_i | X_i)$ satisfy:

$$|\tilde{R}_i - R_i| = |\tilde{F}(Y_i | X_i) - F_{Y|X}(Y_i | X_i)| \leq \varepsilon.$$

Thus, all “tilde” scores lie within ε of their ideal uniform counterparts.

(2) If conformal calibration were applied to the true scores $R_1, \dots, R_n$, which are i.i.d. Uniform(0,1), then setting

$$Q^* = R_{(m)}, \quad \text{where } m = \lceil (1 - \alpha)(n+1) \rceil,$$

Split Conformal Prediction implies the marginal guarantee

$$\mathbb{P}(R_{n+1} \leq Q^*) \geq \frac{m}{n+1} \geq 1 - \alpha.$$

Furthermore, From theorem 1, because the R_i are i.i.d. Uniform(0,1) and thus their distribution is independent of X_i , this guarantee holds conditionally as well:

$$\mathbb{P}(R_{n+1} \leq Q^* \mid X_{n+1} = x) = \mathbb{P}(R_{n+1} \leq Q^*) \geq 1 - \alpha.$$

(3) Since each $\tilde{R}_i$ is within ε of R_i , their m -th order statistics satisfy:

$$Q^* - \varepsilon \leq \tilde{R}_{(m)} = \hat{Q} \leq Q^* + \varepsilon.$$

This follows because shifting all n points in a dataset by at most ε can move any order-statistic by at most ε .

(4) We want to lower-bound the conditional coverage probability:

$$\begin{aligned} \mathbb{P}(Y_{n+1} \in C_n(x) \mid X_{n+1} = x) &= \mathbb{P}(\tilde{F}(Y_{n+1} | x) \leq \hat{Q} \mid x) = \mathbb{P}(\tilde{R}_{n+1} \leq \hat{Q} \mid x) \\ &\geq \mathbb{P}(R_{n+1} \leq \hat{Q} - \varepsilon \mid x) \quad (\text{since } \tilde{R}_{n+1} \leq R_{n+1} + \varepsilon) \\ &\geq \mathbb{P}(R_{n+1} \leq (Q^* - \varepsilon) - \varepsilon \mid x) \quad (\text{since } \hat{Q} \geq Q^* - \varepsilon) \\ &= \mathbb{P}(R_{n+1} \leq Q^* - 2\varepsilon \mid x) \end{aligned}$$

But $R_{n+1} \sim \text{Uniform}(0, 1)$ given x . Hence

$$\begin{aligned} \mathbb{P}(R_{n+1} \leq Q^* - 2\varepsilon \mid x) &= \mathbb{E}_{Q^*} [\mathbb{P}(R_{n+1} \leq Q^* - 2\varepsilon \mid x, Q^*)] \\ &= \mathbb{E}_{Q^*} [(Q^* - 2\varepsilon)_+] \geq \mathbb{E}_{Q^*} [Q^* - 2\varepsilon] \quad (\text{Jensen's inequality}) \\ &= \mathbb{E}_{Q^*} [Q^*] - 2\varepsilon = \frac{m}{n+1} - 2\varepsilon \geq 1 - \alpha - 2\varepsilon. \end{aligned}$$

□

Lemma .1 Let $S_1, \dots, S_T$ and $\tilde{S}_1, \dots, \tilde{S}_T$ be real numbers such that

$$|S_t - \tilde{S}_t| \leq \varepsilon \quad \text{for all } t = 1, \dots, T.$$

And $S_{(1)} \leq \dots \leq S_{(T)}$ and $\tilde{S}_{(1)} \leq \dots \leq \tilde{S}_{(T)}$ be the order statistics of $S_1, \dots, S_T$ and $\tilde{S}_1, \dots, \tilde{S}_T$ respectively. Then for every $m = 1, \dots, T$,

$$|S_{(m)} - \tilde{S}_{(m)}| \leq \varepsilon$$

Proof: There are m, S_t such that $S_t \leq S_{(m)}$, and for each t we have

$$\tilde{S}_t \leq S_t + \varepsilon \leq S_{(m)} + \varepsilon$$

which yields

$$\tilde{S}_{(m)} \leq S_{(m)} + \varepsilon.$$

Doing the same for the lower-bound proves the statement. $\square$

Proposition 3 (Conditional coverage in time-series) Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $F_{t-1} = \sigma(Y_1, \dots, Y_{t-1})$. Suppose we have a correct prior for the conditional distribution with respect to F_{t-1} of Y_t through, $F_t(y) = \mathbb{P}(Y_t \leq y | F_{t-1})$, and define

$$S_t = F_t(Y_t), \quad t = 1, \dots, T+1.$$

Then if split conformal calibration is applied to the transformed values S_i , by taking the order statistic $\hat{Q} = S_{(\lceil (1-\alpha)(T+1) \rceil)}$, the resulting upper prediction band

$$C(F_t) = \{y : F_{t+1}(y) \leq \hat{Q}\}$$

satisfies

$$\mathbb{P}(Y_{T+1} \in C(F_T) \mid F_T) \geq 1 - \alpha.$$

Proof: By the probability-integral transform and correctness of the prior, each $S_t = F_t(Y_t) \sim \text{Unif}(0, 1)$. Furthermore, as the prior is correctly specified a priori, the $\{S_1, \dots, S_{T+1}\}$ are i.i.d. $\text{Uniform}(0, 1)$. Then exchangeability implies the standard split conformal result

$$\mathbb{P}(S_{T+1} \leq \hat{Q}) \geq 1 - \alpha.$$

But since $\hat{Q}$ is a function of $S_1, \dots, S_T$ which are independent of S_{T+1} , this inequality holds conditionally on F_T : $\mathbb{P}(S_{T+1} \leq \hat{Q} \mid F_T) = \mathbb{P}(S_{T+1} \leq \hat{Q})$, so in fact $\mathbb{P}(S_{T+1} \leq \hat{Q} \mid F_T) \geq 1 - \alpha$. Finally, setting $\hat{Q}_Y(F_{t-1}) = F_t^{-1}(\hat{Q})$, so that that $\{Y_{T+1} \leq \hat{Q}_Y(F_T)\} = \{F_{T+1}(Y_{T+1}) \leq \hat{Q}\}$, maps the coverage bound back to the Y -space. Concluding that,

$$\mathbb{P}(Y_{T+1} \in C(F_T) \mid F_T) \geq 1 - \alpha.$$

$\square$

Proposition 4 (Approximate prior in time-series) Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $F_{t-1} = \sigma(Y_1, \dots, Y_{t-1})$. Fix a miscoverage level $\alpha \in (0, 1)$ and let

$$F_t(y) = \mathbb{P}(Y_t \leq y \mid F_{t-1})$$

be the true conditional CDF. Suppose we have an approximate conditional CDF $\tilde{F}_t(y)$, also adapted to F_{t-1} satisfying

$$\sup_{1 \leq t \leq T+1} \sup_{y \in \mathbb{R}} |\tilde{F}_t(y) - F_t(y)| \leq \varepsilon \quad \text{a.s.}$$

Define the non-conformity scores

$$\tilde{S}_i = \tilde{F}_i(Y_i), \quad i = 1, \dots, T.$$

and let

$$m = \lceil (1 - \alpha)(T + 1) \rceil, \quad \hat{Q} = \tilde{S}_{(m)},$$

the m -th order statistic of the $\tilde{S}_i$ over the calibration sample $t = 1, \dots, T$. Then the one-sided conformal prediction set

$$C(F_{t-1}) = \{y \in \mathbb{R} : \tilde{F}_t(y) \leq \hat{Q}\}$$

satisfies, the finite-sample **conditional** coverage bound

$$\mathbb{P}\{Y_{T+1} \in C(F_T) \mid F_T\} \geq 1 - \alpha - 2\varepsilon.$$

Proof:

(1) By assumption, for every t and every realization $Y_1, \dots, Y_t$, the true scores $S_i = F_t(Y_t)$ and the approximate scores $\tilde{S}_t = \tilde{F}_t(Y_t)$ satisfy:

$$|\tilde{S}_t - S_t| = |\tilde{F}_t(Y_t) - F_t(Y_t)| \leq \varepsilon.$$

Thus, all “tilde” scores lie within ε of their ideal uniform counterparts.

(2) If conformal calibration were applied to the true scores $S_1, \dots, S_T$, which are i.i.d. Uniform(0,1), then setting

$$Q^* = S_{(m)}, \quad \text{where } m = \lceil (1 - \alpha)(T + 1) \rceil,$$

Split Conformal Prediction implies the marginal guarantee

$$\mathbb{P}(S_{T+1} \leq Q^*) \geq \frac{m}{T+1} \geq 1 - \alpha.$$

Furthermore, From theorem 1, because the S_t are i.i.d. Uniform(0,1) and thus their distribution is independent of F_{t-1} , this guarantee holds conditionally as well:

$$\mathbb{P}(S_{T+1} \leq Q^* \mid F_T) = \mathbb{P}(S_{T+1} \leq Q^*) \geq 1 - \alpha.$$

(3) Since each $\tilde{S}_t$ is within ε of S_t , their m -th order statistics satisfy:

$$Q^* - \varepsilon \leq \tilde{S}_{(m)} = \hat{Q} \leq Q^* + \varepsilon.$$

This follows because shifting all T points in a dataset by at most ε can move any order-statistic by at most ε . See lemma .1.

(4) We want to lower-bound the conditional coverage probability:

$$\begin{aligned} \mathbb{P}(Y_{T+1} \in C(F_T) \mid F_T) &= \mathbb{P}(\tilde{F}_t(Y_{n+1}) \leq \tilde{Q} \mid F_T) = \mathbb{P}(\tilde{S}_{T+1} \leq \hat{Q} \mid F_T) \\ &\geq \mathbb{P}(S_{T+1} \leq \hat{Q} - \varepsilon \mid F_T) \quad (\text{since } \tilde{S}_{n+1} \leq S_{n+1} + \varepsilon) \\ &\geq \mathbb{P}(S_{T+1} \leq (Q^* - \varepsilon) - \varepsilon \mid F_T) \quad (\text{since } \hat{Q} \geq Q^* - \varepsilon) \\ &= \mathbb{P}(S_{T+1} \leq Q^* - 2\varepsilon \mid F_T) \end{aligned}$$

But $S_{T+1} \sim \text{Uniform}(0, 1)$ given F_T . Hence

$$\begin{aligned} \mathbb{P}(S_{T+1} \leq Q^* - 2\varepsilon \mid F_T) &= \mathbb{E}_{Q^*} [\mathbb{P}(S_{T+1} \leq Q^* - 2\varepsilon \mid F_T, Q^*)] \\ &= \mathbb{E}_{Q^*} [(Q^* - 2\varepsilon)_+] \geq \mathbb{E}_{Q^*} [Q^* - 2\varepsilon] \quad (\text{Jensen's inequality}) \\ &= \mathbb{E}_{Q^*} [Q^*] - 2\varepsilon = \frac{m}{T+1} - 2\varepsilon \geq 1 - \alpha - 2\varepsilon. \end{aligned}$$

□